Date 21/08/26

Expt. No. 3 Page No. 18

## Aim of the experiment:-

Design and implementation of Gray code to Binary code and Binary code to Gray code converters

## Objective:-

1. To design and implement Gray Code to Binary Code and Binary code to Gray code converters
2. To verify the outputs using their respective truth tables.

## Apparatus Required:-

| SL. No. | Name of the component | Specification | Quantity |
|---|---|---|---|
| 1. | Trainer kit (analog and digital) | XPO-ANDIG1 | 1 |
| 2. | Breadboard | – | 1 |
| 3. | X-OR Gate (IC-74LS86) | Quad-2-input XOR Gate | 1 |
| 4. | Connecting wires | – | As per Requirement |

Teacher's Signature

Expt. No. 3 Date 21/08/26

Page No. 17

Theory:-

Gray code is a binary code in which two successive code values differ by only one bit.

1. Gray code to Binary code conversion

In Gray Code to Binary code conversion, the binary equivalent is obtained from the binary equivalent code. The most significant bit (MBS) of the is the same as the MSB of the Gray code. Each subsequent binary bit is obtained by performing the Exclusive-OR (XOR) operation between the previous binary bit and corresponding Gray code bit

2. Binary code to Gray code conversion

In Binary Code to Gray code conversion, the Gray code equivalent is obtained from the binary code. The most significant bit (MSB) of the Gray code is the same as the MSB of the binary code. Each subsequent Gray code bit obtained by performing the Exclusive-OR (X-OR) operation between two adjacent binary bit

Teacher's Signature

# Design and implementation of Gray code to Binary code converter

The Boolean expression are:

$$B_3 = G_3$$
$$B_2 = B_3 \oplus G_2$$
$$B_1 = B_2 \oplus G_1$$
$$B_0 = B_1 \oplus G_0$$

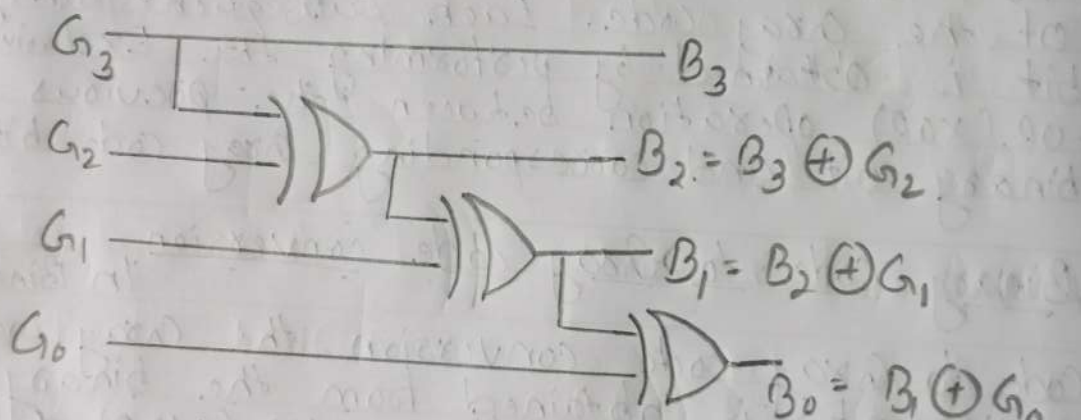



Fig 3.1 : Logic Circuit Diagram of 4-bit Gray Code to Binary code converter

Expt. No. 3

Date 21/08/26

Page No. 18

## Truth Table of Gray code to Binary code

Table 3.1

| Gray code | | | | Binary code | | | |
|---|---|---|---|---|---|---|---|
| MSB | | | LSB | MSB | | | LSB |
| $G_3$ | $G_2$ | $G_1$ | $G_0$ | $B_3$ | $B_2$ | $B_1$ | $B_0$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

Teacher's Signature _____

# Design and Implementation of Binary code to Gray Code Converter

The Boolean expression are:

$$G_3 = B_3$$
$$G_2 = B_3 \oplus B_2$$
$$G_1 = B_2 \oplus B_1$$
$$G_0 = B_1 \oplus B_0$$

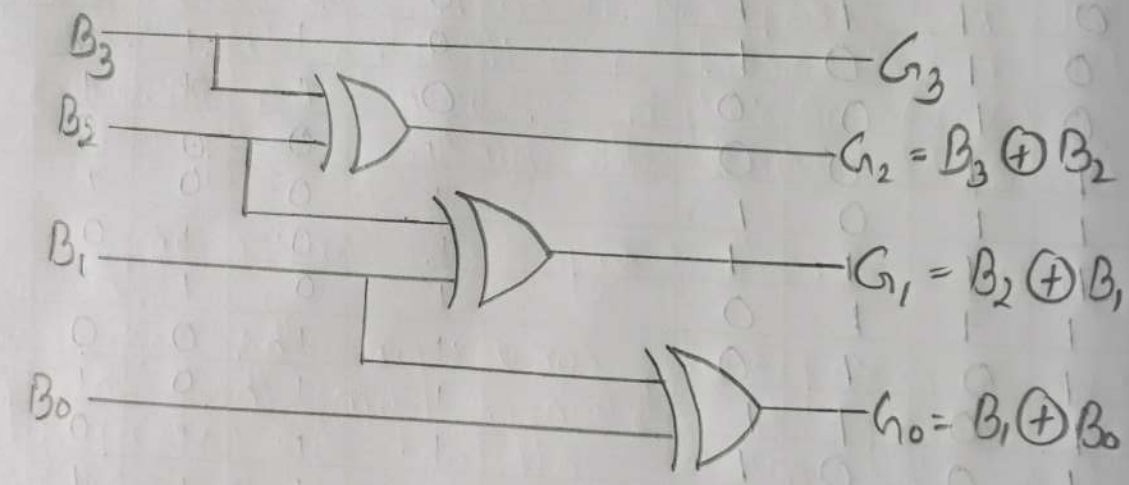



Fig 3.2: Logic Circuit Diagram of 4bit Binary code to Gray code converter

Date 21/08/26

Expt. No. 3 Page No. 19

## Truth Table of Binary code to Gray code

Table 3.2 :-

| Binary code | | | | Gray code | | | |
|---|---|---|---|---|---|---|---|
| MSB | | | LSB | MSB | | | LSB |
| $B_3$ | $B_2$ | $B_1$ | $B_0$ | $G_3$ | $G_2$ | $G_1$ | $G_0$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |

Teacher's Signature

Expt. No. 3 Date 21/08/26

Page No. 20

Procedure :-

(A) Gray code to Binary code converter

1. Connect the required components and power supply to the digital trainer kit
2. Implement the Gray code to Binary code converter circuit according to the Boolean expression
3. Apply different combination of Gray code input to the circuit
4. Observe and record the corresponding Binary code outputs
5. Compare the observed outputs with the theoretical truth table.

(B) Binary Code to Gray Code Converter

1. Connect the required components and power supply to the digital trainer kit.
2. Implement the Binary code to Gray code converter circuit according to the boolean expression.
3. Apply different combination of Binary code input to the circuit.

Teacher's Signature

Expt. No. 3 Date 21/08/26

Page No. 21

4. Observe and record the corresponding Gray code outputs

5. Compare the observed outputs with the theoretical truth table.

## Observation Table :-

1. Observation Table of Gray to Binary code

Observation Table 1 :-

| Input | | | | Output | | | | Remark |
|---|---|---|---|---|---|---|---|---|
| $G_3(D_7)$ | $G_2(D_6)$ | $G_1(D_5)$ | $G_0(D_4)$ | $B_3(7)$ | $B_2(6)$ | $B_1(5)$ | $B_0(4)$ | |
| OFF | OFF | OFF | OFF | NG | NG | NG | NG | |
| OFF | OFF | OFF | ON | NG | NG | NG | G | The |
| OFF | OFF | ON | ON | NG | NG | G | NG | practical |
| OFF | OFF | ON | OFF | NG | NG | G | G | output |
| OFF | ON | ON | OFF | NG | G | NG | NG | match |
| OFF | ON | ON | ON | NG | G | NG | G | the |
| OFF | ON | OFF | ON | NG | G | G | NG | Theo- |
| OFF | ON | OFF | OFF | NG | G | G | G | retical |
| ON | ON | OFF | OFF | G | NG | NG | NG | out- |
| ON | ON | OFF | ON | G | NG | NG | G | put |
| ON | ON | ON | ON | G | NG | G | NG | |
| ON | ON | ON | OFF | G | NG | G | G | |
| ON | OFF | ON | OFF | G | G | NG | NG | |
| ON | OFF | ON | ON | G | G | NG | G | |
| ON | OFF | OFF | ON | G | G | G | NG | |
| ON | OFF | OFF | OFF | G | G | G | G | |

(ON-1, OFF-0) (NG- Not Glowing, G- Glowing)

Teacher's Signature

Date 21/08/25

Expt. No. 3

Page No. 22

2. Observation Table of Binary code to Gray code.

Observation Table 2:-

| Input | | | | output | | | | |
|---|---|---|---|---|---|---|---|---|
| $B_3$ | $B_2$ | $B_1$ | $B_0$ | $G_3$ | $G_2$ | $G_1$ | $G_0$ | Remark |
| ($D_7$) | ($D_6$) | ($D_5$) | $D_4$ | (7) | (6) | (5) | 4 | |
| OFF | OFF | OFF | OFF | NG | NG | NG | NG | |
| OFF | OFF | OFF | ON | NG | NG | NG | G | |
| OFF | OFF | ON | OFF | NG | NG | G | G | |
| OFF | OFF | ON | ON | NG | NG | G | NG | |
| OFF | ON | OFF | OFF | NG | G | G | NG | The practical output matches the theoretical output |
| OFF | ON | OFF | ON | NG | G | G | G | |
| OFF | ON | ON | OFF | NG | G | NG | G | |
| OFF | ON | ON | ON | NG | G | NG | NG | |
| ON | OFF | OFF | OFF | G | G | NG | NG | |
| ON | OFF | OFF | ON | G | G | NG | G | |
| ON | OFF | ON | OFF | G | G | G | G | |
| ON | OFF | ON | ON | G | G | G | NG | |
| ON | ON | OFF | OFF | G | NG | G | NG | |
| ON | ON | OFF | ON | G | NG | G | G | |
| ON | ON | ON | OFF | G | NG | NG | G | |
| ON | ON | ON | ON | G | NG | NG | NG | |

(ON-1, OFF-0) (NG-Not Glowing, G-Glowing)

Teacher's Signature

Date 21/08/26

Expt. No. 3 Page No. 23

Conclusion:-

The Gray code to Binary code and Binary code to Gray code converter circuits have been successfully implemented using the respective Boolean expressions. Different combination of input codes have been applied and corresponding output codes have been obtained and recorded. The observed outputs have matched the theoretical truth Tables.

Name- Sambit Kumar Jena
Regd no- 2511O283
Branch- Computer Engineering
Group- A2
Sub Group- 2

Teacher's Signature